Course outline (Syllabus of the course)

Course Code: CS3234	**Course Name: Foundations of Generative AI**	
Semester: 6	**Area:** Specialization	**SEE Type**: Theory (30 Marks)
Credits: 3	**No. of hours per week (L + T + P):** 2 + 0 + 2	**Contact Hrs:** 60 Hours

Prerequisite(Course/Skill/Knowledge):

- CS2213 – Introduction to Machine Learning, CS2820 – Calculus for Computer Science
- Programming Skills (Python)

Syllabus:

Module - 1 Introduction to Generative AI	**No. of hours**
	5
Introduction to Generative AI and its applications, Introduction to prompt engineering, Tokenization, embeddings, context length LLMs vs. SLMs – architectures and trade-offs Introduction to Gen AI Tools - Lang Chain, Hugging Face, Databricks, Lang Flow, Kaggle, Fast AI, Vector DBs	

Module - 2 Principles of Prompt Engineering	**No. of hours**
	5
Principles for designing effective prompts (such as Persona Pattern, Root Prompts, RGC, Chain of Thought, Comparative Prompting). Techniques for controlling the style, tone, and content of generated text, strategies for incorporating external knowledge into prompts, approaches to handle complex or multi-part prompts, Prompt Injection	

Module - 3 Introduction to Generative Models	**No. of hours**
	5
Evolution of generative models–Generative adversarial networks (GAN), Other Variants of GANs such as DCGAN, StyleGAN, Cycle GAN, Autoencoders, Variable Autoencoders (VAEs) -KL Divergence, Stable Diffusion and Denoising techniques. Diffusion Models vs. GANs comparison.	

Module - 4 Evolution of Generative Pre-Trained Models	**No. of hours**

	10

Limitations of RNNs/LSTM, Attention mechanisms and Transformer Encoder-Decoder Architecture, Encoder only model e.g. BERT. Decoder only, Autoregressive models e.g. GPT. Self-supervised training, Fine tuning of LLMs and its various techniques, Multimodal LLM Architecture considerations, Implementation of Retrieval-Augmented Generation (RAG) architecture, Graph RAG. Introduction to LLM Agents, Tools and Frameworks for Building Agents (such as Lang Chain, Phi Data)

Module - 5 Applications of LLMs and Ethical Considerations	**No. of hours**
	5

Using multimodal LLMs to generate text, audio, image and video contents.
Using LLMs for NLP, computer vision, software development tasks (such as code generation, bug detection and correction, code refactoring, code translation). Using Cursor AI, GitHub Copilot for software projects
Ethical Principles in LLM, Bias Detection and Mitigation Techniques, Data privacy and Security, Fairness, Transparency, Guard Rail, and Accountability, Regulatory Compliance (EU AI Act, India's DPDP Bill) including privacy, user consent, user control, and user feedback mechanisms in LLMs

Course outcomes

1. Explore fundamental concepts of Generative AI and its diverse applications in various industries
2. Apply prompt engineering principles when working with LLMs such as ChatGPT
3. Utilize multimodal LLMs to generate text, audio, image, video content
4. Apply LLMs into software engineering workflows such as code generation, refactoring, and debugging
5. Apply generative AI skills to develop a project that addresses a real-world use case

Text Books

1	"Build a Large Language Model from Scratch" by Sebastian Raschka (Author) Manning Publications, 2024, ISBN-13978-1633437166
2	"Hands-On Large Language Models: Language Understanding and Generation" 1st Edition by Jay Alammar, Maarten Grootendorst, 2024, Shroff/O'Reilly ISBN-13 - 978-9355425522
3	"LLM Engineer's Handbook" by Paul Iusztin (Author), Maxime Labonne Packt Publishing, ISBN-13 - 978-1836200079
4	"Large Language Models: a Deep Dive: Bridging Theory and Practice" by Uday Kamath, Kevin Keenan, Garrett Somers, Sarah Sorenson 2024, Springer-Nature New York Inc, ISBN-13 - 978-3031656460

Reference books	
1	“Designing Machine Learning Systems" by Chip Huyen, O'Reilly Media, 2024. ISBN-13 - 978-9355422675
2	Generative Deep Learning, David Foster, O'Reilly Media, 2023, ISBN-978-1098134181

Lab Programs (30 Hours)	
1	First prompts using ChatGPT/Gemini/Perplexity/Claude, Prompt example of launch Email for a Workshop, Using DALL-E via Image creator, creating a video using Fliki.ai and Kalaido.ai
2	Using Kaggle, Fast.AI, Hugging Face (Installation of Transformer Module, Sentiment analysis, NLP, Translation and summarization and sentence embedding)
3	Use of Persona Prompt, Use of Cognitive Verifier Pattern, Question Refinement Pattern, Provide New Information and Ask Questions, Root Prompt
4	Practice Chain of thought, Tabular format, fill in the blank, RGC, zero shot, one shot and few shots prompting
5	Using ChatGPT/Gemini: Generate your resume, Simulate a complete interview
6	Using Cursor AI/Lovable: Create a website, web app
7	Gemini-Pro: Generate API Key, Examples for accessing the Model using API Key
8	Using Meta's LLama3 Models using Replicate/Groq / Use Ollama/LM Studio to deploy Llama3 locally
9	Experimenting open-source models with hugging face (zero shot audio classification, automatic speech recognition, Text to speech)
10	Hands on lab using Lang Chain and Lang Flow
11	Implement RAG based LLM Project using Vector DB
12	Build a Simple LLM Agent using phi data framework
13	Use LLMs for Code Generation and Bug Detection in Software Development
14	Using multimodal LLMs for Visual Question Answering
15	Implement a course project using LLMs such as Llama/Gemini with stream lit / gradio as front-end

Lesson plan

Name of the Programme:	**B.Tech (Hons)**	
Title of the Course:	**Foundations of Generative AI**	
Course code	**CS3234**	
Semester:	**VI**	
Course credit	**No. of hours per week** **(2+0+2)**	**Total no. of Teaching hours**
3		**60 Hours**

Session	Module No.	Topic	Pedagogy /activities	Pre-reading/reference
1,2	1	Introduction to Generative AI and its applications, Introduction to prompt engineering. Overview of Large Language Models (LLMs) and Small Language Models (SLMS) and their architectures	Active learning with Powerpoint presentation and in class activities	Build a Large Language Model from Scratch by Sebastian Raschka
3,4		Tokenization, embeddings, context length LLMs vs. SLMs	Active learning with Powerpoint presentation and in class activities	Build a Large Language Model from Scratch by Sebastian Raschka
5,6		**Lab Ex. 10:** Hands on lab using Lang Chain and Lang Flow Hugging Face, Data Bricks, FastAI, Vector Databases	**Hands-on Practice in Lab**	Refer Python Coding
7,8		**Lab Ex.2:** Using Kaggle, Fast.AI, Hugging Face (Installation of Transformer Module, Sentiment analysis, NLP, Translation and summarization and sentence embedding)	**Hands-on Practice in Lab**	Refer Python Coding

9,10		**Lab Ex.1:** First prompts using ChatGPT/Gemini/Perplexity/Claude, Prompt example of launch Email for a Workshop, Using DALL-E via Image creator, creating a video using Fliki.ai and Kalaido.ai	**Hands-on Practice in Lab**	Refer Python Coding
11		Word2Vector and Simple Applications in Gen AI	**Hands-on in Classroom**	Refer Python Coding
12,13	**2**	**Lab Ex. 3:** Use of Persona Prompt, Use of Cognitive Verifier Pattern, Question Refinement Pattern, Provide New Information and Ask Questions, Root Prompt	**Hands-on Practice in Lab**	Refer Python Coding
14		Techniques for controlling the style, tone, and content of generated text, strategies for incorporating external knowledge into prompts, Approaches to handle complex or multi-part prompts	Active learning with Powerpoint presentation and in class activities	Build a Large Language Model from Scratch by Sebastian Raschka
15		Approaches to handle complex or multi-part prompts, Prompt Injection	Active learning with Powerpoint presentation and in class activities	Build a Large Language Model from Scratch by Sebastian Raschka
16,17		**Lab Ex.4:** Practice Chain of thought, Tabular format, fill in the blank, RGC, zero shot, one shot and few shots prompting	**Hands-on Practice in Lab**	Refer Python Coding
18		Evolution of generative models–Generative adversarial networks (GAN)	Active learning with Powerpoint presentation and in class activities	Build a Large Language Model from Scratch by Sebastian Raschka
19	**3**	Other Variants of GANs such as DCGAN, StyleGAN, Cycle GAN	Active learning with Powerpoint presentation and in class activities	Build a Large Language Model from Scratch by Sebastian Raschka

20	3	Autoencoders, Variable Autoencoders (VAEs) -KL Divergence	Active learning with Powerpoint presentation and in class activities	Build a Large Language Model from Scratch by Sebastian Raschka
21		Stable Diffusion and De-noising techniques. Diffusion Models vs. GANs comparison.	Active learning with Powerpoint presentation and in class activities	Build a Large Language Model from Scratch by Sebastian Raschka
22,23		**Lab Ex. 5:** Using ChatGPT/Gemini: Generate your resume, Simulate a complete interview	**Hands-on Practice in Lab**	Refer Python Coding
24,25		**Lab Ex. 6:** Using Cursor AI/Lovable: Create a website, web app	**Hands-on Practice in Lab**	Refer Python Coding
26	4	Limitations of RNNs/LSTM, Attention mechanisms and Transformer	Active learning with Powerpoint presentation and in class activities	Build a Large Language Model from Scratch by Sebastian Raschka
27		Encoder only model e.g. BERT. Decoder only, Autoregressive models e.g. GPT.	Active learning with Powerpoint presentation and in class activities	Build a Large Language Model from Scratch by Sebastian Raschka
28		Self-supervised training, Fine tuning of LLMs and its various techniques	Active learning with Powerpoint presentation and in class activities	Build a Large Language Model from Scratch by Sebastian Raschka
29,30		**Lab Ex. 11:** Implement RAG based LLM Project using Vector DB	**Hands-on Practice in Lab**	Refer Python Coding
31,32		**Lab Ex. 12:** Build a Simple LLM Agent using phi data framework	**Hands-on Practice in Lab**	Refer Python Coding

33,34		**Lab Ex. 7:** Gemini-Pro: Generate API Key, Examples for accessing the Model using API Key	**Hands-on Practice in Lab**	Refer Python Coding
35,36		**Lab Ex. 8:** Using Meta's LLama3 Models using Replicate/Groq / Use Ollama/LM Studio to deploy Llama3 locally	**Hands-on Practice in Lab**	Refer Python Coding
37,38	**5**	**Lab Ex. 14:** Using multimodal LLMs for Visual Question Answering	**Hands-on Practice in Lab**	Refer Python Coding
39,40		**Lab Ex. 13:** Use LLMs for Code Generation and Bug Detection in Software Development	**Hands-on Practice in Lab**	Refer Python Coding
41		Using Cursor AI, GitHub Copilot for software projects	Active learning with Powerpoint presentation and in class activities	Build a Large Language Model from Scratch by Sebastian Raschka
42		Ethical Principles in LLM, Bias Detection and Mitigation Techniques	Active learning with Powerpoint presentation and in class activities	Build a Large Language Model from Scratch by Sebastian Raschka
43		Data privacy and Security, Fairness, Transparency, Guard Rail, and Accountability, Regulatory Compliance (EU AI Act, India's DPDP Bill) including privacy, user consent, user control, and user feedback mechanisms in LLMs	Active learning with Powerpoint presentation and in class activities	Build a Large Language Model from Scratch by Sebastian Raschka
44		Experimenting open-source models with hugging face (zero shot audio classification, automatic speech recognition, Text to speech)	Active learning with Powerpoint presentation and in class activities	Build a Large Language Model from Scratch by Sebastian Raschka
45,46		**Lab Ex. 15:** Implement a course project using LLMs such as Llama/Gemini with stream lit / gradio as front-end	**Hands-on Practice in Lab**	Refer Python Coding

47,48	**Tutorial and Demo Videos**	Practical Execution Applications using Gen AI in real-time	**Hands-on Demo**	Refer Python Coding
49,50	**Tutorial and Demo Videos**	Practical Execution Neural Networks and applications	**Hands-on Demo**	Refer Python Coding
51,52	**Tutorial and Demo Videos**	Practical Execution CNN and applications	**Hands-on Demo**	Refer Python Coding
53,54	**Tutorial and Demo Videos**	Databricks Demo in class	**Hands-on Demo**	Refer Python Coding
55,56	**Tutorial and Demo Videos**	Practical Execution of Gen AI in Underwater Imaging for simple examples	**Hands-on Demo**	Refer Python Coding
57,58	**Tutorial and Demo Videos**	Practical Execution of Gen AI in Stock Market Trading	**Hands-on Demo**	Refer Python Coding
59,60	**Tutorial and Demo Videos**	Practical Execution of Gen AI in Supply Chain Management	**Hands-on Demo**	Refer Python Coding

Assessment and Evaluation

Assessment Component	Type of Component	Rubrics for Assessment
1	CIE 1 Test MOOC, In-Class Activities	**Component 1** – CIE 1 Online Quiz – 10 Marks **Component 2** – MOOC, In-Class Activities (10 Marks)
2	CIE 2 – Pen and Paper Exam	**Component 3** – Test will be conducted for **25 marks** and will be evaluated according to Scheme of Evaluation prepared by team of course faculty
3	CIE 3 - Practical Component	**Component 4** - Course Project (20 Marks) Lab Record Submission (5 Marks)

Semester End Exams (SEE): 30%	
Mode of Exam	SEE
Weightage of exam	30%

Course Outcomes- Progamme Outcomes Matrix

Course Outcomes		PO 1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	Explore fundamental concepts of Generative AI and its diverse applications in various industries	3	2	2	2	3	-	-	-	2	2	-	2
C02	Apply prompt engineering principles when working with LLMs such as ChatGPT	2	2	2	1	3	-	-	-	2	2	-	2
C03	Utilize multimodal LLMs to generate text, audio, image, and video content	3	2	2	2	3	-	-	-	2	2	-	2
CO4	Apply generative AI skills to develop a project that addresses a real-world use case	3	2	2	2	3	-	-	-	3	3	2	2
CO5	Apply generative AI skills to develop a project that addresses a real-world use case	2	2	2	1	3	-	-	-	2	2	-	2

Rubrics for evaluation of CIE (Consolidated Table CIE component)

Internal Assessment Plan: 70 Marks				
Sl#	**Component**	**Marks**	**Type of Assessment**	**Timeline**
CIE 1 (20 Marks)				
1	CO1	**10**	**Component 1- Online Quiz**	Week 3
2	CO1, CO2	**10**	**Component 2 - MOOC, In-Class Activities**	Week 5
CIE 2 (25 Marks)				
3	CO1 - CO3	**25**	**Component 3 – Pen and Paper Exam**	Week 9
CP 3 (25 Marks) [For practical subjects to be conducted out of 50 reduce to 25]				
4	CO1 – CO4	**CIE 3 - Graded Component 3**		
		20	**Component 4 – Course Project**	Week 13
		5	**Lab Record Submission**	Week 14
		25		

Rubrics for evaluation of CIE (separate for each CIE component)

Total Marks		**20**
CIE-1 Components		**Rubrics for Assessment**
CIE-1	Quiz Graded Component 1	Quiz will be conducted for 10 marks. Each right answer carries one mark each. No negative marking for wrong answers.
	MOOC Courses: Graded Component-2	Certificate of completion for 10 marks. Students should complete MOOC Courses and submit the proof in Google Classroom. ***(Refer Table-1)***

Table-1:

Graded Component-2 (10 Marks):

Assess students' ability to complete MOOC courses and reflect on their learning.

Criteria	Excellent (10 Marks)	Satisfactory (5 Marks)	Needs Improvement (0 Marks)
Completion of Courses (10 Marks)	Successfully completes all MOOC Courses before the deadline. Certificates are uploaded to Google Classroom with no errors.	Completes some courses or has minor errors in uploading certificates.	Fails to complete the courses or does not upload certificates.

Total Marks		**25**
CIE-2 Components		**Rubrics for Assessment**
CIE-2	Theory	Test will be conducted for 25 marks and will be evaluated according to Scheme of Evaluation prepared by team of course faculty.

Practical Component (CIE 3)	Rubrics for Assessment
a) **Course Project**	
Viva	10 Marks
Demo	5 Marks
PPT	5 Marks
Total – 20 Marks	
b) **Lab Record Submission**	**5 Marks**
Total (a+b)	**25 Marks**

Table-2

Lab Record Soft Copy and Execution rubrics **(Max: 5 marks)**					
Sl. No	Criteria	Measuring Methods	Excellent (2 Marks)	Good (1 Mark)	Poor (0.5 Marks)
1.	Understanding of problem statements. (1 Marks)	Observations	Student exhibits a thorough understanding of the Gen AI requirements and selects an appropriate algorithm.	Student exhibits a thorough understanding of the Gen AI requirements and selects an appropriate algorithm.	Student exhibits a thorough understanding of the Gen AI requirements and selects an appropriate algorithm.
2.	Execution (2 Marks)	Observations	Student exhibits a thorough understanding of the Gen AI requirements and selects an appropriate algorithm.	Student exhibits a thorough understanding of the Gen AI requirements and selects an appropriate algorithm.	Student exhibits a thorough understanding of the Gen AI requirements and selects an appropriate algorithm.

3.	Results and Documentatio n (2 Marks)	Observations	Student exhibits a thorough understanding of the Gen AI requirements and selects an appropriate algorithm.	Student exhibits a thorough understanding of the Gen AI requirements and selects an appropriate algorithm.	Student exhibits a thorough understanding of the Gen AI requirements and selects an appropriate algorithm.
Viva Voce Rubrics (Max: 10 marks)					
Sl. No	Criteria	Measuring Methods	Excellent (2 Marks)	Good (1 Mark)	Poor (0.5 Marks)
1.	Conceptual Understandin g of Gen AI Techniques (5 Marks)	Viva Voce	Thoroughly explains the GEN AI algorithms and related concepts.	Adequately explains the Gen AI algorithms and related concepts.	Unable to explain Gen AI concepts, algorithms and related concepts.
2.	Application of GEN AI Problem-Solv ing Techniques (5 Marks)	Viva Voce	Insightfully explains appropriate Gen AI problem-solvi ng techniques for the given problem.	Sufficiently explains the use of Gen AI problem-solvin g techniques for the given problem.	Unable to explain the Gen AI problem-solving techniques for the given problem.

Table-3

Project Demo Rubrics **(Max: 5 marks)**					
Sl. No	Criteria	Measuring Methods	Excellent (2 Marks)	Good (1 Mark)	Poor (0.5 Marks)
1.	Understanding of problem statements. (1 Marks)	Observations	Student exhibits a thorough understanding of the Gen AI requirements and selects an appropriate algorithm.	Student exhibits a thorough understanding of the Gen AI requirements and selects an appropriate algorithm.	Student exhibits a thorough understanding of the Gen AI requirements and selects an appropriate algorithm.
2.	Execution (2 Marks)	Observations	Student exhibits a thorough understanding of the Gen AI requirements and selects an appropriate algorithm.	Student exhibits a thorough understanding of the Gen AI requirements and selects an appropriate algorithm.	Student exhibits a thorough understanding of the Gen AI requirements and selects an appropriate algorithm.
3.	Results and Documentation (2 Marks)	Observations	Student exhibits a thorough understanding of the Gen AI requirements and selects an appropriate algorithm.	Student exhibits a thorough understanding of the Gen AI requirements and selects an appropriate algorithm.	Student exhibits a thorough understanding of the Gen AI requirements and selects an appropriate algorithm.

Table-4

Presentation PPT Rubrics **(Max: 5 marks)**					
Sl. No	Criteria	Measuring Methods	Excellent (2 Marks)	Good (1 Mark)	Poor (0.5 Marks)
1.	Understanding of problem statements. (1 Marks)	Observations	Student exhibits a thorough understanding of the Gen AI requirements and selects an appropriate algorithm.	Student exhibits a thorough understanding of the Gen AI requirements and selects an appropriate algorithm.	Student exhibits a thorough understanding of the Gen AI requirements and selects an appropriate algorithm.
2.	Execution (2 Marks)	Observations	Student exhibits a thorough understanding of the Gen AI requirements and selects an appropriate algorithm.	Student exhibits a thorough understanding of the Gen AI requirements and selects an appropriate algorithm.	Student exhibits a thorough understanding of the Gen AI requirements and selects an appropriate algorithm.
3.	Results and Documentation (2 Marks)	Observations	Student exhibits a thorough understanding of the Gen AI requirements and selects an appropriate algorithm.	Student exhibits a thorough understanding of the Gen AI requirements and selects an appropriate algorithm.	Student exhibits a thorough understanding of the Gen AI requirements and selects an appropriate algorithm.

Signature of the Course Faculty